Year 10 Higher

REVISION PACK

Progress Check 1

Thursday 1 October 2026

Name: _____

Class: _____

Units in this pack

- Number 1
- Algebra 1

Number 1 · Working with fractions (Lesson 1.1)

Worked example: adding fractions

Work out $\frac{3}{4} + \frac{5}{6}$

Give your answer as a mixed number.

Step 1	Same bottom	$\frac{9}{12} + \frac{10}{12}$
Step 2	Add the tops	$\frac{19}{12}$
Step 3	Answer	$1\frac{7}{12}$

Practice: adding fractions

1.1 Work out $\frac{4}{5} + \frac{2}{3}$. Give your answer as a mixed number.

1.2 Work out $\frac{5}{6} + \frac{3}{8}$. Give your answer as a mixed number.

1.3 Work out $\frac{3}{4} + \frac{2}{3}$. Give your answer as a mixed number.

1.4 Work out $\frac{5}{7} + \frac{1}{2}$. Give your answer as a mixed number.

1.5 Work out $\frac{7}{8} + \frac{1}{3}$. Give your answer as a mixed number.

1.6 Work out $\frac{5}{9} + \frac{5}{6}$. Give your answer as a mixed number.

Number 1 · Working with fractions (Lesson 1.1)

Worked example: multiplying mixed numbers

Work out $1\frac{2}{3} \times 2\frac{1}{4}$

Give your answer as a mixed number in its simplest form.

Step 1 Improper fractions	$\frac{5}{3} \times \frac{9}{4}$
Step 2 Multiply	$\frac{45}{12}$
Step 3 Simplify	$\frac{15}{4}$
Step 4 Answer	$3\frac{3}{4}$

Practice: multiplying mixed numbers

1.7 Work out $1\frac{1}{2} \times 2\frac{2}{3}$

1.8 Work out $3\frac{1}{3} \times 1\frac{3}{5}$. Give your answer as a mixed number in its simplest form.

1.9 Work out $2\frac{1}{4} \times 1\frac{5}{6}$. Give your answer as a mixed number in its simplest form.

1.10 Work out $1\frac{3}{4} \times 2\frac{2}{5}$. Give your answer as a mixed number in its simplest form.

1.11 Work out $2\frac{2}{3} \times 1\frac{1}{8}$

1.12 Work out $3\frac{3}{4} \times 1\frac{2}{3}$. Give your answer as a mixed number in its simplest form.

Number 1 · Working with fractions (Lesson 1.1)

Worked example: dividing mixed numbers

Work out $2\frac{1}{4} \div 1\frac{1}{2}$

Step 1 Improper fractions	$\frac{9}{4} \div \frac{3}{2}$
Step 2 Flip and multiply	$\frac{9}{4} \times \frac{2}{3} = \frac{18}{12}$
Step 3 Simplify	$\frac{3}{2}$
Step 4 Answer	$1\frac{1}{2}$

Practice: dividing mixed numbers

1.13 Work out $4\frac{1}{2} \div 1\frac{1}{5}$. Give your answer as a mixed number in its simplest form.

1.14 Work out $2\frac{2}{3} \div 1\frac{1}{3}$

1.15 Work out $3\frac{3}{4} \div 2\frac{1}{2}$. Give your answer as a mixed number in its simplest form.

1.16 Work out $5\frac{1}{4} \div 1\frac{3}{4}$

1.17 Work out $2\frac{2}{5} \div 1\frac{1}{5}$

1.18 Work out $4\frac{1}{6} \div 1\frac{2}{3}$. Give your answer as a mixed number in its simplest form.

Number 1 · Working with fractions (Lesson 1.1)

Worked example: subtracting with a mixed number

Work out $2\frac{1}{4} - \frac{2}{3}$

Step 1 Improper fraction	$\frac{9}{4} - \frac{2}{3}$
Step 2 Same bottom	$\frac{27}{12} - \frac{8}{12}$
Step 3 Subtract the tops	$\frac{19}{12}$
Step 4 Answer	$1\frac{7}{12}$

Practice: subtracting with a mixed number

1.19 Work out $2\frac{1}{3} - \frac{5}{6}$

1.20 Work out $1\frac{3}{5} - \frac{3}{4}$

1.21 Work out $3\frac{1}{4} - \frac{2}{3}$

1.22 Work out $2\frac{2}{5} - \frac{5}{6}$

1.23 Work out $1\frac{5}{6} - \frac{3}{4}$

1.24 Work out $2\frac{1}{2} - \frac{4}{5}$

Number 1 · Working with fractions (Lesson 1.1)

Worked example: fraction problems

Sara spends $\frac{1}{3}$ of the money on rent. Sara spends $\frac{1}{4}$ of the rest on food. What fraction of the money does Sara have left?

Step 1	The rest	$1 - \frac{1}{3} = \frac{2}{3}$
Step 2	Food	$\frac{1}{4} \times \frac{2}{3} = \frac{1}{6}$
Step 3	Left	$\frac{2}{3} - \frac{1}{6} = \frac{1}{2}$
Step 4	Answer	$\frac{1}{2}$

Practice: fraction problems

1.25 Omar spends $\frac{2}{5}$ of the money on rent. Omar spends $\frac{1}{3}$ of the rest on food. What fraction of the money does Omar have left?

1.26 Huda spends $\frac{1}{4}$ of the money on clothes. Huda spends $\frac{2}{3}$ of the rest on books. What fraction of the money does Huda have left?

1.27 Yousef spends $\frac{3}{8}$ of the money on a phone. Yousef spends $\frac{1}{5}$ of the rest on games. What fraction of the money does Yousef have left?

1.28 Mona spends $\frac{1}{6}$ of the money on food. Mona spends $\frac{3}{5}$ of the rest on travel. What fraction of the money does Mona have left?

1.29 Fahad spends $\frac{3}{7}$ of the money on rent. Fahad spends $\frac{1}{2}$ of the rest on food. What fraction of the money does Fahad have left?

1.30 Laila spends $\frac{2}{9}$ of the money on books. Laila spends $\frac{3}{7}$ of the rest on clothes. What fraction of the money does Laila have left?

Number 1 · Order of operations and rounding (Lessons 1.2, 1.3)

Worked example: order of operations

This question is about the order of operations.

(a) Work out $2 + 4 \times 3^2$

(b) Insert one pair of brackets to make this calculation correct.
$8 + 2 \times 5 - 1 = 49$

(a)	**Step 1**	Indices first	$3^2 = 9$
	Step 2	Then multiply	$4 \times 9 = 36$
	Step 3	Then add	$2 + 36 = 38$
(b)	**Step 1**	Brackets round 8 + 2	$(8 + 2) \times 5 - 1$
	Step 2	Check	$10 \times 5 - 1 = 49$

Practice: order of operations

2.1 This question is about the order of operations.

(a) Work out $7 + 2 \times 5^2$

(b) Insert one pair of brackets to make this calculation correct. $3 + 5 \times 4 - 2 = 30$

2.2 This question is about the order of operations.

(a) Work out $20 - 3 \times 2^3$

(b) Insert one pair of brackets to make this calculation correct. $2 \times 6 + 3 - 1 = 17$

2.3 This question is about the order of operations.

(a) Work out $6 + 18 \div 3^2$

(b) Insert one pair of brackets to make this calculation correct. $4 + 8 \times 2 + 1 = 28$

2.4 This question is about the order of operations.

(a) Work out $9 + 4 \times 2^3$

(b) Insert one pair of brackets to make this calculation correct. $5 \times 4 - 1 + 2 = 17$

2.5 This question is about the order of operations.

(a) Work out $3 + 6 \times 2^2$

(b) Insert one pair of brackets to make this calculation correct. $10 - 2 \times 3 + 1 = 25$

2.6 This question is about the order of operations.

(a) Work out $40 - 5 \times 3^2$

(b) Insert one pair of brackets to make this calculation correct. $6 + 2 \times 7 - 3 = 53$

Number 1 · Order of operations and rounding (Lessons 1.2, 1.3)

Worked example: rounding

Round 562.738

(a) correct to 1 decimal place. (b) correct to 2 significant figures.

(a)	**Step 1**	Next digit is 3: round down	562.7
(b)	**Step 1**	Next digit is 2: round down	560

Practice: rounding

2.7 Round 275.846

(a) correct to 1 decimal place. (b) correct to 2 significant figures.

2.8 Round 1093.472

(a) correct to 1 decimal place. (b) correct to 2 significant figures.

2.9 Round 48.0961

(a) correct to 1 decimal place. (b) correct to 2 significant figures.

2.10 Round 736.258

(a) correct to 1 decimal place. (b) correct to 2 significant figures.

2.11 Round 5281.63

(a) correct to 1 decimal place. (b) correct to 2 significant figures.

2.12 Round 19.954

(a) correct to 1 decimal place. (b) correct to 2 significant figures.

Algebra 1 · Substitution and simplifying (Lessons 2.1, 2.2, 2.3)

Worked example: substitution into v = u + at

$v = u + at$. Work out the value of v when $u = 12$, $a = -2$ and $t = 5$

Step 1	Put the numbers in	$v = 12 + (-2) \times 5$
Step 2	Multiply first	$= 12 - 10$
Step 3	Answer	$v = 2$

Practice: substitution into v = u + at

3.1 $v = u + at$. Work out the value of v when $u = 3$, $a = -4$ and $t = 2$

3.2 $v = u + at$. Work out the value of v when $u = 10$, $a = -3$ and $t = 6$

3.3 $v = u + at$. Work out the value of v when $u = -4$, $a = 5$ and $t = 3$

3.4 $v = u + at$. Work out the value of v when $u = 20$, $a = -6$ and $t = 3$

3.5 $v = u + at$. Work out the value of v when $u = -2$, $a = -3$ and $t = 4$

3.6 $v = u + at$. Work out the value of v when $u = 15$, $a = -2.5$ and $t = 4$

Algebra 1 · Substitution and simplifying (Lessons 2.1, 2.2, 2.3)

Worked example: substitution with a square

$E = \frac{1}{2}mv^2$. Work out the value of E when $m = 2$ and $v = -7$

Step 1	Put the numbers in	$E = \frac{1}{2} \times 2 \times (-7)^2$
Step 2	Square first	$= \frac{1}{2} \times 2 \times 49$
Step 3	Answer	$E = 49$

Practice: substitution with a square

3.7 $E = \frac{1}{2}mv^2$. Work out the value of E when $m = 6$ and $v = -5$

3.8 $E = \frac{1}{2}mv^2$. Work out the value of E when $m = 10$ and $v = -2$

3.9 $E = \frac{1}{2}mv^2$. Work out the value of E when $m = 8$ and $v = -3$

3.10 $E = \frac{1}{2}mv^2$. Work out the value of E when $m = 4$ and $v = -6$

3.11 $E = \frac{1}{2}mv^2$. Work out the value of E when $m = 12$ and $v = -1$

3.12 $E = \frac{1}{2}mv^2$. Work out the value of E when $m = 3$ and $v = -4$

Algebra 1 · Substitution and simplifying (Lessons 2.1, 2.2, 2.3)

Worked example: simplifying expressions

Simplify

(a) $6p + 2q - 4p + 3q$ (b) $2a^2b \times 5ab^3$

(a)	**Step 1**	Collect p	$6p - 4p = 2p$
	Step 2	Collect q	$2q + 3q = 5q$
	Step 3	Answer	$2p + 5q$
(b)	**Step 1**	Multiply the numbers	$2 \times 5 = 10$
	Step 2	Add the powers	$a^{2+1}b^{1+3} = a^3b^4$
	Step 3	Answer	$10a^3b^4$

Practice: simplifying expressions

3.13 Simplify

(a) $7x + 4y - 3x - y$ (b) $4a^3b \times 2ab^2$

3.14 Simplify

(a) $5m - 2n + m + 6n$ (b) $3xy^2 \times 6x^2y$

3.15 Simplify

(a) $9c + 3d - 4c - 7d$ (b) $5p^2q^2 \times 2pq^4$

3.16 Simplify

(a) $8x - 3y + 2x + 7y$ (b) $6a^2b^3 \times 3ab$

3.17 Simplify

(a) $4m + 9n - 7m - 2n$ (b) $2x^3y \times 7x^2y^2$

3.18 Simplify

(a) $3p - 5q + 4p + 8q$ (b) $5ab^2 \times 4a^3b$

Worked example: unknowns on both sides

Solve $7x - 4 = 3x + 16$

Step 1	Subtract $3x$	$4x - 4 = 16$
Step 2	Add 4	$4x = 20$
Step 3	Divide by 4	$x = 5$

Practice: unknowns on both sides

4.1 Solve $6x - 5 = 2x + 11$

4.2 Solve $9x + 4 = 5x + 24$

4.3 Solve $8x - 3 = 5x + 12$

4.4 Solve $7x + 2 = 3x - 10$

4.5 Solve $10x - 9 = 4x + 15$

4.6 Solve $5x + 8 = 2x + 23$

Worked example: equations with fractions

Solve $\dfrac{3x-1}{4}=5$

Step 1	Multiply by 4	$3x-1=20$
Step 2	Add 1	$3x=21$
Step 3	Divide by 3	$x=7$

Practice: equations with fractions

4.7 Solve $\dfrac{2x+5}{3}=7$

4.8 Solve $\dfrac{4x-3}{5}=5$

4.9 Solve $\dfrac{5x+2}{2}=6$

4.10 Solve $\dfrac{3x+4}{5}=2$

4.11 Solve $\dfrac{2x-1}{3}=5$

4.12 Solve $\dfrac{6x+3}{3}=5$

Worked example: equations with brackets

Solve $2(3x+1) = 5(x+2)$

Step 1	Expand	$6x + 2 = 5x + 10$
Step 2	Subtract $5x$	$x + 2 = 10$
Step 3	Subtract 2	$x = 8$

Practice: equations with brackets

4.13 Solve $4(2x-1) = 3(x+7)$

4.14 Solve $5(x-2) = 2(2x+3)$

4.15 Solve $3(3x+2) = 4(x+9)$

4.16 Solve $2(4x-3) = 3(x+3)$

4.17 Solve $6(x-1) = 4(x+2)$

4.18 Solve $4(2x+1) = 2(3x+1)$